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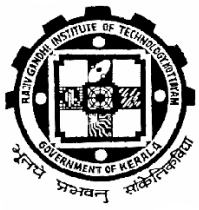
DISASTER RESPONSE AND MONITORING ROVER

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BACHELOR OF TECHNOLOGY
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CERTIFICATE

This is to certify that the Project Report entitled **DISASTER RESPONSE AND MONITORING ROVER** is a bonafide record of the work done by **Abhitha Francis (KTE23EC009), Aiswarya N L (KTE23EC016), Arjun N K (KTE23EC026), Shilpa Ann Varghese (KTE23EC062)** under our guidance towards the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering** of A P J Abdul Kalam Technological University, during the year 2025-26.

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ABSTRACT

The increasing demand for efficient surveillance and rapid response in disaster-prone and hazardous environments has driven the development of intelligent robotic systems. This project presents the design and implementation of a ground-based disaster surveillance rover intended to assist rescue teams and law enforcement agencies in monitoring and operating within inaccessible or high-risk areas. The rover is equipped with an integrated sensing system comprising a camera module and Passive Infrared (PIR) sensors to continuously monitor the surroundings and detect the presence of human beings .

A key feature of the system is its ability to detect and classify human presence based on their condition, distinguishing between individuals in motion or not in motion. Based on this classification, the system generates and transmits appropriate alerts to facilitate timely and informed rescue operations. In addition to human detection, the rover incorporates a hazard detection module capable of identifying toxic gas leakage and abnormal temperature variations, thereby enhancing situational awareness and safety.

The system is designed to provide real-time alerts and environmental data to a control unit, enabling rapid decision-making during emergency scenarios. By integrating multiple sensing technologies with intelligent detection mechanisms, the proposed rover offers a low-cost, scalable, and efficient solution for disaster management and surveillance applications. Experimental evaluation demonstrates that the system effectively improves response time, operational safety, and monitoring capability in critical environments, making it a promising tool for modern rescue and surveillance missions.

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Chapter 1

INTRODUCTION

Disaster and hazardous environments require efficient monitoring systems to ensure safety and quick response. Manual operations in such conditions are often risky and sometimes impossible due to inaccessibility and the presence of dangers like toxic gases and extreme temperatures.

To overcome these challenges, robotic systems such as ground-based surveillance rovers are used for safe and reliable monitoring. These systems can navigate terrains and provide real-time information without risking human lives.

This project presents the design and implementation of a disaster and surveillance rover equipped with a camera and sensors to detect human presence. The system provides real-time alerts, improving safety and efficiency in rescue and surveillance operations.

1.1 OBJECTIVES

The main objective of this project is to design and implement a ground-based disaster surveillance rover capable of operating in hazardous and inaccessible environments. The system aims to develop an efficient monitoring framework using a combination of sensors and a camera module to detect human presence in real time. The rover is designed to accurately identify and classify human conditions, distinguishing between human in motion and not in motion, and generate appropriate alerts to assist rescue teams in decision-making.

Furthermore, the project focuses on integrating hazard detection mechanisms to identify toxic gas leakage and abnormal temperature variations, ensuring improved safety during disaster situations. Wireless communication is incorporated

to enable real-time data transmission and remote monitoring from a control unit. Overall, the project strives to achieve a low-cost, scalable, and reliable robotic system that enhances efficiency, safety, and responsiveness in surveillance and rescue operations.

1.2 MOTIVATION

Despite advancements in disaster management technologies, many existing surveillance and rescue systems are either expensive or limited in functionality. Manual rescue operations in hazardous environments expose personnel to significant risks, including toxic gases, extreme temperatures, and unstable structures. Additionally, the lack of real-time monitoring and intelligent detection systems can delay response and reduce the chances of successful rescue.

This gap between safety, efficiency, and affordability motivated the development of a disaster and surveillance rover. By integrating embedded systems, sensor fusion, and cost-effective components, the project aims to provide a reliable and accessible solution for monitoring hazardous environments. The proposed system enhances situational awareness and supports rescue teams by reducing human risk and improving response effectiveness.

1.3 SCOPE OF THE PROJECT

The scope of this project includes the design, development, and testing of a ground-based disaster surveillance rover capable of operating in hazardous and inaccessible environments. It involves the integration of multiple sensors such as Passive Infrared (PIR) sensors, gas sensors, and temperature sensors along with a camera module for real-time monitoring and data acquisition. The system performs continuous detection of human presence, enabling efficient situational analysis.

The rover is designed to process sensor data and identify human conditions, distinguishing between human in motion or not in motion, and generate appropriate

alerts. Additionally, hazard detection mechanisms are incorporated to identify toxic gas leakage and abnormal temperature variations, ensuring improved safety during rescue operations. Wireless communication is implemented to enable real-time data transmission and remote control through a user interface.

Through the integration of embedded systems, sensor fusion, and intelligent detection mechanisms, the project aims to deliver a cost-effective, reliable, and scalable robotic system that enhances efficiency, safety, and responsiveness in disaster management and surveillance applications.

Chapter 2

LITERATURE REVIEW

Vimal et al. [1] designed and implemented an IoT-enabled rover using the ESP32 microcontroller and ESP32-CAM module for real-time object detection, environmental monitoring, and web-based control. The system integrates ultrasonic sensors for obstacle detection and sensors such as DHT11 and MQ-2 for monitoring temperature, humidity, and gas concentration. The ESP32-CAM utilizes the YOLO algorithm to perform real-time object detection, enabling the identification of humans, animals, and vehicles. Additionally, the rover supports remote operation through a web-based interface using HTTP protocol, allowing users to monitor live video feeds and sensor data. The system demonstrated good performance in real-time detection and environmental data acquisition with relatively low cost and power consumption. However, the use of lightweight hardware like ESP32 limits computational capability, which may affect detection accuracy and model complexity. Furthermore, reliance on HTTP communication may introduce latency, and the system may face challenges in highly dynamic or large-scale disaster environments.

Pragatheeswari et al. [2] proposed an IoT-integrated smart rover system designed for disaster relief management with a focus on environmental monitoring. The rover is equipped with multiple sensors such as temperature, humidity, gas, flame, and vibration sensors, enabling real-time data acquisition and hazard detection. The system uses ESP32 and Blynk platform for remote monitoring, live video streaming, and alert generation, enhancing situational awareness and decision-making during disaster scenarios. Despite its advantages in environmental sensing and remote operation, the system lacks advanced human detection capabilities and relies primarily on sensor data rather than intelligent vision-based recognition, limiting its effective-

ness in locating survivors.

Angalaeswari et al. [3] developed a multi-sensor based rover system incorporating a rocker-bogie mechanism for enhanced mobility in uneven terrains. The system utilized an Arduino microcontroller integrated with a wide range of sensors, including temperature, humidity, gas, flame, ultrasonic, IR, sound, and touch sensors, to monitor environmental conditions and detect potential hazards. Wireless communication through Bluetooth or Wi-Fi enabled real-time data transmission and remote control of the rover. The design emphasized continuous environmental monitoring and robustness in harsh conditions, making it suitable for disaster-prone areas. However, the system relied primarily on sensor-based detection and lacked advanced vision-based or AI-driven human detection capabilities. Additionally, the use of numerous sensors increased system complexity and power consumption, while the absence of autonomous decision-making limited its efficiency in dynamic real-world rescue scenarios.

Roshan et al. [4]developed a deep learning-based human detection rover for search-and-rescue operations in disaster zones. The system integrates PIR sensors with a YOLOv3 object detection model running on a Raspberry Pi to identify human presence in real time. It also incorporates GPS tracking and wireless communication through WiFi and MQTT for transmitting location data to rescue teams. The system demonstrated high accuracy and real-time responsiveness while maintaining low cost and power consumption. However, the reliance on PIR-triggered detection may miss stationary victims, and performance can be affected in extremely low-light or highly cluttered environments.

Dhanya et al. [5] introduced an intelligent rescue rover utilizing transfer learning and embedded systems for autonomous human detection in disaster environments. The system employs SSD MobileNet architecture for detecting human body parts using a high-resolution camera and processes data on a Raspberry Pi 5. It also

integrates LiDAR, thermal imaging, GPS, and long-range communication modules to enhance navigation and detection accuracy. The rover uses a rocker-bogie mechanism for improved mobility on rough terrains and supports autonomous navigation with obstacle avoidance. Although the system provides high detection accuracy and adaptability, it requires higher computational resources and faces challenges related to power consumption, system complexity, and reliability in highly dynamic environments.

Chapter 3

SYSTEM DESCRIPTION

The Disaster and Surveillance Rover is designed for monitoring and rescue operations in hazardous and inaccessible environments. It integrates sensors and a camera module to provide real-time surveillance, detect human presence, and identify abnormal environmental temperature and presence of toxic gases. The rover is suitable for disaster zones such as collapsed buildings, fire-affected areas, and toxic gas leaks. The system comprises five main subsystems: sensing, control and processing, mobility, communication, and power supply. The sensing unit includes PIR sensors for motion detection, gas sensors for hazardous leaks, and temperature sensors for abnormal heat patterns. A camera module captures live video for remote monitoring. The control unit, based on an ESP32 microcontroller, collects and processes sensor data to trigger alerts and guide movement. It enables manual navigation, differentiating between humans, obstacles, and hazards. Alerts and video streams are transmitted wirelessly to a remote control interface for real-time operation.
The mobility system uses DC motors and a motor driver to navigate uneven terrain, turn, and reverse efficiently. A rechargeable battery powers the system, ensuring portability and extended operation.

Overall, the Disaster Surveillance Rover provides a cost-effective, scalable, and efficient solution for improving situational awareness, response times, and safety in disaster management and surveillance scenarios. Its modular design allows future integration of additional sensors or robotic attachments for more advanced operations.

3.1 BLOCK DIAGRAM

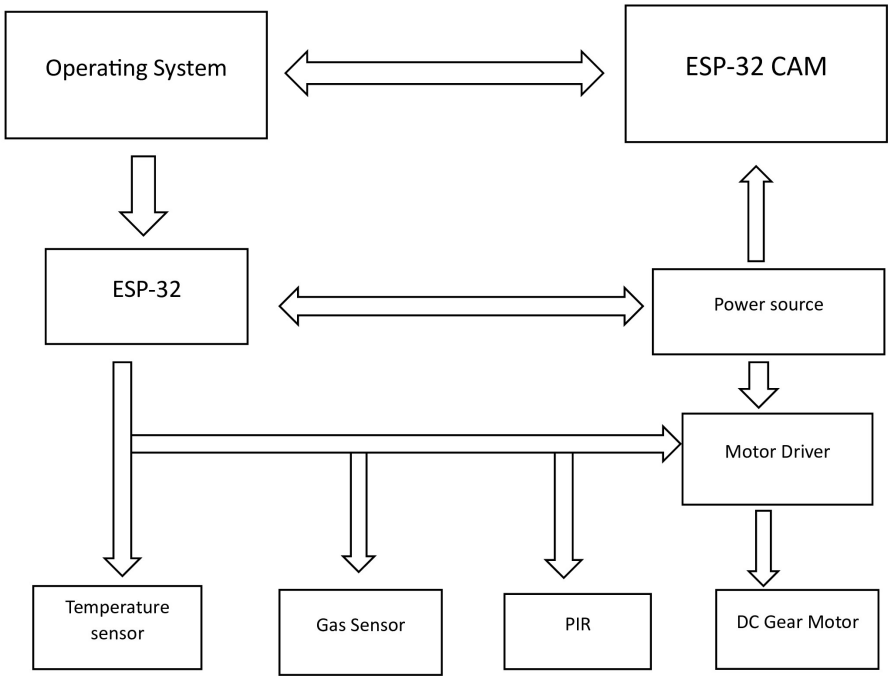


Figure 3.1: **Block Diagram**

The block diagram of the system is shown in Figure 3.1. It illustrates the main functional modules of the Disaster Surveillance Rover: sensing, control, mobility, communication, and power management. Sensors, including PIR, gas, temperature sensors, and a camera, detect human presence. The ESP32 microcontroller processes the sensor data and generates control signals for the DC motors, enabling navigation. Alerts and live video are transmitted to a remote interface for real-time monitoring. The rechargeable battery powers all modules, ensuring portable and continuous operation. The system works in a closed-loop manner to provide efficient disaster surveillance and rescue support.

3.2 WORKING PRINCIPLE

The Disaster and Surveillance Rover operates by continuously monitoring its environment using sensors and a camera module. PIR, gas, and temperature sensors

detect human presence and hazardous conditions, while the camera provides live video for situational awareness. The ESP32 microcontroller processes the sensor data, filters noise, and determines appropriate actions. Control signals are sent to the DC motors, enabling navigation. Simultaneously, the communication module transmits alerts and live video to a remote interface for real-time monitoring. The system operates in a closed-loop manner, allowing detection of hazards and humans, and providing reliable support for disaster surveillance and rescue operations.

3.3 FUNCTIONAL OVERVIEW

The Disaster Surveillance Rover operates through an integrated sequence of sensing, processing, actuation, and communication functions. The overall operation can be divided into five main stages — sensing, data processing, decision making, actuation, and alerting — which interact in real time to ensure effective disaster monitoring and rescue support.

The sensing unit, comprising PIR, gas, and temperature sensors along with a camera, continuously monitors the environment for human presence. The analog and digital data from these sensors are transmitted to the ESP32 microcontroller, which filters, interprets, and classifies the information to determine appropriate actions.

Based on the command from the user, control unit generates signals to the DC motors for navigation. Simultaneously, alerts and live video are sent to the remote control interface via Wi-Fi, enabling real-time monitoring and decision-making. The rechargeable battery powers all subsystems, ensuring continuous operation in field conditions. This closed-loop operation allows the rover to respond manually, improving situational awareness, safety, and efficiency in disaster and surveillance operations.

3.4 MAIN FUNCTIONS

Sensor Monitoring and Data Acquisition:

The rover is equipped with PIR, gas, and temperature sensors, along with a camera module. These sensors continuously detect human presence, hazardous conditions, and environmental parameters. Sensor data is collected and sent to the ESP32 microcontroller for processing.

Data Processing and Decision Making:

The ESP32 filters and interprets sensor signals in real time. Based on predefined thresholds and rules, it identifies human presence, or hazardous conditions and determines appropriate responses.

Navigation and Actuation:

Control signals from the ESP32 drive the DC motors via a motor driver, enabling the rover to move, turn, and reverse efficiently. Motion of the rover is manual, that is it is controlled by the user.

Communication and Alerts:

The live video feed and alert notifications are transmitted to a remote interface through Wi-Fi, allowing real-time monitoring and decision-making by operators. This ensures timely response during disaster or surveillance operations.

Power Management:

The rover is powered by a rechargeable battery pack. Efficient power regulation ensures continuous operation of the sensors, motors, microcontroller, and communication modules.

The Disaster and Surveillance Rover integrates sensing, processing, mobility, and communication in a compact, modular design. It provides a cost-effective, reliable, and scalable solution for disaster monitoring, rescue operations, and surveillance in hazardous environments.

Chapter 4

SOFTWARES USED

4.1 ARDUINO IDE

Arduino IDE is an open-source integrated development environment widely used for programming microcontrollers such as ESP32. It provides a simple and efficient platform for writing, compiling, and uploading embedded C/C++ code. The environment supports a wide range of libraries that simplify hardware interfacing, making it highly suitable for rapid prototyping of IoT-based systems.

In this project, Arduino IDE is used to program both the ESP32 and ESP32-CAM modules. It facilitates the integration of various sensors including the PIR motion sensor, LM35 temperature sensor, and MQ135 gas sensor. The software enables real-time data acquisition, processing of sensor inputs, and execution of control logic. Additionally, it is used to configure WiFi communication between the ESP32 and the laptop, allowing seamless data transmission and remote control.

The Arduino environment also manages motor control through the L298N motor driver by generating appropriate digital signals for directional movement of the rover. Furthermore, it supports HTTP communication required for sending alert messages via the Telegram Bot API. Overall, Arduino IDE serves as the backbone for embedded system control and hardware interaction in the project.

4.2 PYTHON

Python is a high-level, versatile programming language widely used in artificial intelligence, data processing, and web development. Its extensive ecosystem of libraries and frameworks makes it an ideal choice for integrating multiple functionalities

within a single application.

In this project, Python plays a central role in implementing the control and processing layer on the laptop. It is used to handle real-time video processing, decision-making logic,rover movement and communication with the ESP32 module. Python scripts manage the execution of the human detection algorithm, manual rover movement and interpret the results to generate appropriate alerts and control signals.

The language also facilitates integration with networking protocols, enabling communication between the web interface and the ESP32 over WiFi.

4.3 VISUAL STUDIO CODE

Visual Studio Code (VS Code) is a lightweight yet powerful source code editor developed by Microsoft. It provides a comprehensive development environment with support for multiple programming languages, debugging tools, and integrated terminal access.

In this project, VS Code is used as the primary development platform for writing and managing all project files, including HTML, CSS, JavaScript, and Python scripts. Its features such as syntax highlighting, code completion, and extension support enhance productivity and simplify the development process.

The integrated terminal in VS Code allows efficient execution of Python scripts and server commands, making it easier to test and debug different components of the system within a single environment.

4.4 ULTRALYTICS YOLOV8

Ultralytics YOLOv8 is a state-of-the-art deep learning model designed for real-time object detection and pose estimation. It is known for its high accuracy and efficiency, making it suitable for applications requiring fast and reliable detection.

In this project, the YOLOv8s-pose model is employed to perform human

detection from the live video stream transmitted by the ESP32-CAM. The model processes each video frame to identify human presence and analyze body posture, enabling intelligent monitoring of disaster environments. This capability enhances the rover's functionality by providing visual confirmation of human presence in addition to sensor-based detection.

The integration of YOLOv8 with Python allows real-time inference and seamless communication with other system components. The use of this advanced computer vision model significantly improves the reliability and effectiveness of the monitoring system.

4.5 OPENCV

OpenCV (Open Source Computer Vision Library) is a widely used library for real-time image and video processing. It provides efficient tools for capturing, processing, and analyzing visual data.

In this project, OpenCV is used to capture the live video stream from the ESP32-CAM and convert it into frames suitable for processing by the YOLOv8 model. It acts as an interface between the camera hardware and the deep learning algorithm, ensuring smooth and continuous video analysis.

Additionally, OpenCV is utilized for basic image preprocessing tasks such as resizing and frame handling, which are essential for optimizing the performance of the detection model. Its real-time processing capability makes it a crucial component in the vision subsystem of the rover.

4.6 FLASK

Flask is a lightweight web application framework written in Python, used for developing web-based interfaces and APIs. It provides a simple yet powerful platform for building interactive applications with minimal overhead.

In this project, Flask is used to develop a web-based control interface for

the rover. The application allows users to manually control the rover's movement and initiate the human detection process through a browser. It acts as a bridge between the user and the hardware system by sending control commands from the web interface to the ESP32 via WiFi.

The Flask server also integrates with the Python-based detection system, enabling coordinated operation of motor control and computer vision tasks. Its flexibility and ease of deployment make it an ideal choice for implementing the control dashboard of the system.

4.7 JAVASCRIPT

JavaScript is a versatile programming language used to create dynamic and interactive web applications. It plays a crucial role in enabling communication between the user interface and the backend system through asynchronous HTTP requests.

In this project, JavaScript is used to handle user interactions such as button clicks for starting the video stream, initiating human detection, and controlling the rover manually. It sends requests to the backend server and processes responses to update the user interface accordingly. This allows real-time control and monitoring through the web dashboard.

The use of JavaScript ensures a responsive and interactive experience, enabling seamless integration between the frontend interface and backend processing components.

4.8 TELEGRAM BOT API

Telegram Bot API is a cloud-based messaging interface that enables applications to send and receive messages through Telegram. It is widely used for implementing real-time notification systems.

In this project, the Telegram Bot API is used to send alerts to users based on sensor readings and detection results. Notifications such as "human detected,"

"motion detected," "gas level exceeded," and "temperature threshold exceeded" are transmitted instantly to the user's Telegram account.

The integration is achieved using HTTP requests from both the ESP32 . This ensures that critical information is delivered in real time, enhancing the effectiveness of the disaster monitoring system.

4.9 HTML AND CSS

HTML (HyperText Markup Language) and CSS (Cascading Style Sheets) are standard technologies used for designing and structuring web pages.

In this project, HTML and CSS are used to create the front-end interface of the web-based control system. The interface includes buttons and controls for rover navigation and system operation. CSS is used to enhance the visual appearance and responsiveness of the web page, ensuring a user-friendly experience.

The combination of HTML and CSS provides a simple yet effective platform for interacting with the rover through a browser, completing the overall system interface.

Chapter 5

HARDWARE DESCRIPTION

The Disaster Response and Monitoring Rover integrates multiple hardware components that collectively perform environmental sensing, real-time monitoring, locomotion, and communication. Each component is selected to ensure efficient operation, reliability, and cost-effectiveness while maintaining system modularity.

The primary hardware components used in this project include the ESP32 development board, ESP32-CAM module, PIR motion sensor, LM35 temperature sensor, MQ135 gas sensor, L298N motor driver, and DC motors. These components work together to enable real-time data acquisition, processing, and remote-controlled navigation of the rover.

5.1 ESP32-CAM MODULE

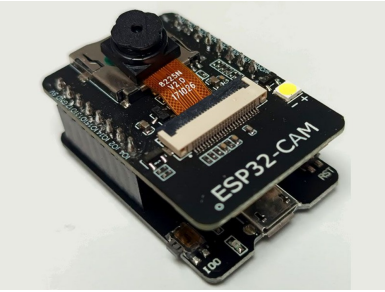


Figure 5.1: ESP32-CAM MODULE

The ESP32-CAM is a compact microcontroller module equipped with an OV2640 camera, capable of capturing and streaming video over WiFi. It combines image acquisition and wireless transmission in a single low-cost module.

In this project, the ESP32-CAM is used to stream live video from the rover to a laptop, where it is processed using a YOLOv8-based human detection algorithm.

This enables real-time visual monitoring of disaster environments and enhances situational awareness.

The module operates independently of the main ESP32 board and communicates over the same network. Its small size and low power consumption make it suitable for integration into mobile robotic platforms.

5.2 ESP32 DEVELOPMENT BOARD



Figure 5.2: ESP32 DEVELOPMENT BOARD

The ESP32 is a low-cost, low-power microcontroller developed by Espressif Systems, featuring integrated Wi-Fi and Bluetooth capabilities. It is powered by a dual-core Tensilica Xtensa LX6 processor operating at up to 240 MHz and provides multiple GPIO pins, as well as analog and digital interfaces suitable for sensor integration.

In this project, the ESP32 serves as the central control unit of the rover. It reads inputs from the PIR sensor, LM35 temperature sensor, and MQ135 gas sensor, processes the data according to predefined thresholds, and generates control signals for the DC motors to navigate and avoid obstacles. The ESP32 also transmits real-time sensor readings and video feeds to a laptop or mobile interface over Wi-Fi,

enabling remote monitoring and operation.

The ESP32 is preferred over traditional microcontrollers due to its high processing capability, built-in wireless communication, and ability to handle multiple tasks concurrently, making it ideal for IoT-enabled robotic systems like the Disaster and Surveillance Rover.

5.3 PIR MOTION SENSOR



Figure 5.3: PIR MOTION SENSOR

The Passive Infrared (PIR) sensor is used to detect motion by measuring changes in infrared radiation emitted by objects in its field of view. It is widely used in security and surveillance systems due to its reliability and low power consumption.

In this project, the PIR sensor detects human movement in the environment. When motion is detected, the sensor outputs a digital HIGH signal to the ESP32. This information is combined with the results from the YOLO-based human detection system to improve the accuracy of human presence detection.

The PIR sensor plays a crucial role in distinguishing between static and moving humans, enabling more informative alert messages.

5.4 LM35 TEMPERATURE SENSOR

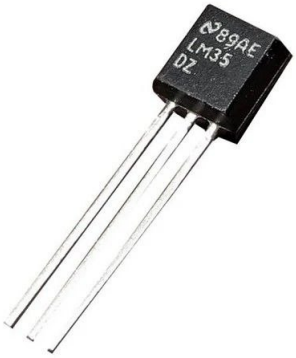


Figure 5.4: LM35 TEMPERATURE SENSOR

The LM35 is a precision analog temperature sensor that provides an output voltage linearly proportional to the temperature in degrees Celsius. It does not require external calibration and offers high accuracy with low self-heating.

In this project, the LM35 sensor is used to monitor ambient temperature in the disaster environment. The analog output is read by the ESP32's ADC, converted into temperature values, and compared with predefined thresholds.

If the temperature exceeds safe limits, an alert message is sent via the Telegram Bot, enabling real-time monitoring of hazardous conditions such as fire or overheating.

5.5 MQ135 GAS SENSOR



Figure 5.5: MQ135 GAS SENSOR

The MQ135 gas sensor is used for detecting harmful gases such as ammonia, nitrogen oxides, benzene, smoke, and carbon dioxide. It operates based on changes in resistance when exposed to different gas concentrations.

In this project, the MQ135 sensor is used to monitor air quality and detect the presence of hazardous gases in disaster zones. The sensor provides an analog output that is read by the ESP32 and evaluated against threshold levels.

Although the MQ135 does not provide highly precise measurements, it is effective for detecting relative changes in gas concentration, making it suitable for early warning applications.

5.6 L298N MOTOR DRIVER

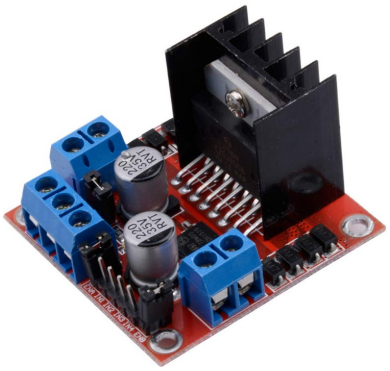


Figure 5.6: L298N MOTOR DRIVER

The L298N motor driver is a dual H-bridge driver capable of controlling the speed and direction of DC motors. It allows bidirectional control and supports high current loads required for motor operation.

In this project, the L298N motor driver is used to control four DC motors connected to the rover wheels. The ESP32 sends control signals to the driver, which regulates motor movement based on user commands received via the web interface.

The driver enables forward, backward, left, and right movements of the rover, making it suitable for navigation in complex environments.

5.7 DC MOTORS



Figure 5.7: DC MOTORS

DC motors are electromechanical devices that convert electrical energy into mechanical motion. In this project, four DC geared motors are used to drive the rover.

These motors provide sufficient torque for movement across uneven surfaces typically encountered in disaster environments. They are controlled via the L298N motor driver, which allows precise direction and speed control.

The use of four motors ensures better stability, load distribution, and maneuverability of the rover.

5.8 LI-ION BATTERY



Figure 5.8: LI-ION BATTERY

Lithium-ion (Li-ion) batteries are widely used in portable electronic systems due to their high energy density, lightweight nature, and rechargeable capability. They provide a stable and efficient power source for embedded and robotic applications.

In this project, a Li-ion battery pack is used as the primary power supply for the rover. A 2-cell (2S) configuration is typically used, providing a nominal voltage of 7.4 V, which is suitable for driving the DC motors through the motor driver.

The battery enables wireless and mobile operation of the rover, eliminating the need for external power sources. However, proper power management is required to ensure stable operation of all components.

5.9 BUCK CONVERTER



Figure 5.9: BUCK CONVERTER

A buck converter is a DC-DC step-down converter used to reduce voltage efficiently while maintaining a stable output. It is commonly used in embedded systems to supply regulated voltage to sensitive components.

In this project, the buck converter steps down the 7.4 V supply from the Li-ion battery to a stable 5 V output. This regulated voltage is used to power the ESP32, ESP32-CAM, and various sensors, ensuring reliable operation.

The use of a buck converter prevents voltage fluctuations that could lead to microcontroller resets or communication failures, thereby improving the overall stability and performance of the system.

Chapter 6

RESULTS

The implementation of the Disaster Surveillance Rover was evaluated through various modules to verify its performance in real-time monitoring and rescue scenarios.



Figure 6.1: ROVER

The ESP32-CAM module successfully streamed live video to the laptop over Wi-Fi with minimal delay, enabling continuous observation of the environment. The captured video frames were processed using the YOLOv8 model, which accurately detected human presence under normal lighting conditions. The integration of the PIR sensor with the detection system improved reliability by distinguishing between moving and non-moving humans, thereby reducing false alerts.

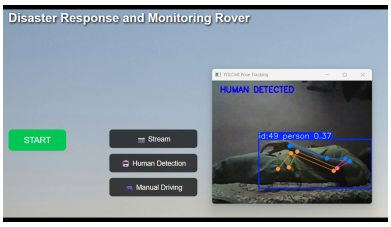


Figure 6.2: HUMAN DETECTION

Environmental monitoring using the LM35 temperature sensor and MQ135 gas sensor provided stable and measurable outputs. Threshold-based evaluation

enabled the system to generate alerts when abnormal temperature or gas levels were detected. These alerts were successfully transmitted to the user through the Telegram platform, ensuring real-time notification.

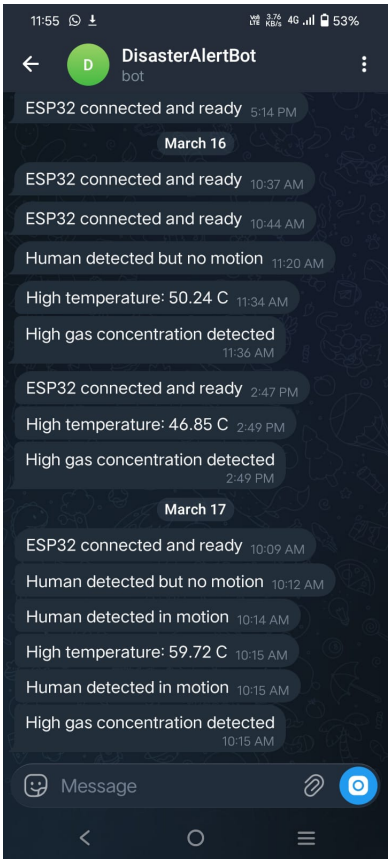


Figure 6.3: TELEGRAM ALERTS

The rover control system, implemented using a Flask-based web interface, demonstrated reliable performance in executing movement commands such as forward, backward, left, and right, as well as initiating live video streaming, thereby enabling smooth, responsive navigation and real-time monitoring.

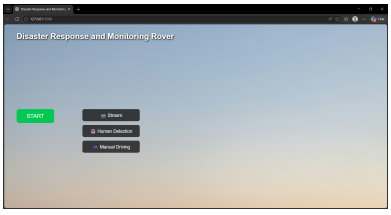


Figure 6.4: WEB INTERFACE

The overall system demonstrated successful integration of video streaming,

human detection, environmental sensing, alert generation, and wireless control. Although minor issues such as voltage fluctuations and sensor sensitivity were observed, the system performed reliably under normal conditions. The results confirm that the proposed rover is a feasible and effective solution for disaster monitoring and surveillance applications.

Chapter 7

CONCLUSION

The Disaster Response and Monitoring Rover is developed as a cost-effective robotic system for assisting in disaster scenarios through real-time monitoring, human detection, and environmental sensing. It enables remote operation in hazardous environments, improving safety and situational awareness.

The system integrates IoT, wireless communication, and computer vision technologies. The ESP32 and ESP32-CAM modules provide data acquisition and live video streaming, while the YOLOv8 model accurately detects human presence. The combination of vision-based detection with PIR sensor input improves reliability by distinguishing between moving and non-moving humans.

Environmental monitoring using LM35 and MQ135 sensors enables detection of temperature variations and changes in air quality. A threshold-based alert system, integrated with the Telegram Bot API, ensures real-time notifications. The Flask-based web interface allows smooth remote control of the rover over Wi-Fi.

The system demonstrates effective coordination between hardware and software components, confirming its feasibility for disaster monitoring applications. Although minor limitations such as network dependency and power fluctuations exist, the prototype performs reliably.

Overall, the project provides a strong foundation for future enhancements such as improved power management, autonomous navigation, and advanced communication systems, making it a promising solution for real-world disaster response.

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